ABSTRACT:

This project uses Blockchain technology to make the agricultural supply chain transparent and trustworthy. Every step—from the farmer to the consumer—is recorded securely, allowing people to trace product origin, ensure quality, and prevent fraud while promoting fair pricing and accountability.

OBJECTIVES:

1. **To reduce supply chain delays** by providing a unified and transparent digital system.
2. **To help farmers receive fair pricing** through clear visibility of product flow.
3. **To minimize wastage** by monitoring product conditions and movements accurately.
4. **To support better decision-making** with real-time, trustworthy data.
5. **To increase overall efficiency** of agricultural supply chain operations through automation and decentralization.

METHODS:

Blockchain Ledger Creation- Using a decentralized ledger to store all supply chain records securely.

Smart Contracts- Automating transactions like product transfer, payment release, and verification without third-party involvement.

QR Code- Tagging each product or batch with QR to track movement from farm to consumer.

Data Recording at Each Stage- Farmers, transporters, warehouses, and retailers update data (quantity, quality, location, time).

LITERATURE SURVEY

Sl. No.	Research Paper	Authors & Year	Key Technologies & Methods	Limitations
1	Blockchain-Based Traceability for Agricultural Products: A Systematic Literature Review	Guangjie Lv in 2023	Systematic review, content Analysis	Lack of real- world implementation.
2	Blockchain Technology in Agriculture: Ensuring Transparency and Traceability	Adebunmi Okechukwu Adewusi in 2023	Blockchain, Smart Contracts, Conceptual Framework.	Lacks practical deployment studies.
3	Survey on Evaluation of Blockchain -Based Agricultural Traceability Systems	Pang S in 2024	Comparative Analysis,Security& Performance Metrics.	Limited system diversity ,absence of unified benchmarks.

PROBLEM STATEMENT:

The agricultural supply chain lacks transparency, leading to food safety issues, inefficiencies, and distrust among stakeholders..

SCOPE OF PROJECT:

1. Review blockchain-based traceability systems in agriculture.
2. Identify technologies and methods used for traceability.
3. Evaluate advantages, challenges, and limitations.
4. Compare existing models and approaches.
5. Suggest improvements and future research directions.

SOFTWARE & HARDWARE REQUIREMENTS:

SOFTWARE:

1. Html
2. Node.js
3. CSS
4. Javascript
5. Bootstrap
6. Mongo DB

HARDWARE:

1. Camera
2. Phone / laptop

Functional & Non-Functional Requirements:

Functional Requirements:

1. **User Authentication:**
   Allows user to register and create accounts.
2. **Produce Registration :**
   Enable farmers to register their produce with unique identifiers (e.g., QR codes).
3. **Tracking and Tracing :**
   Allow users to track and trace produce throughout the supply chain.
4. **Reporting :**
   Generate reports on produce movement, quality, and other relevant metrics.

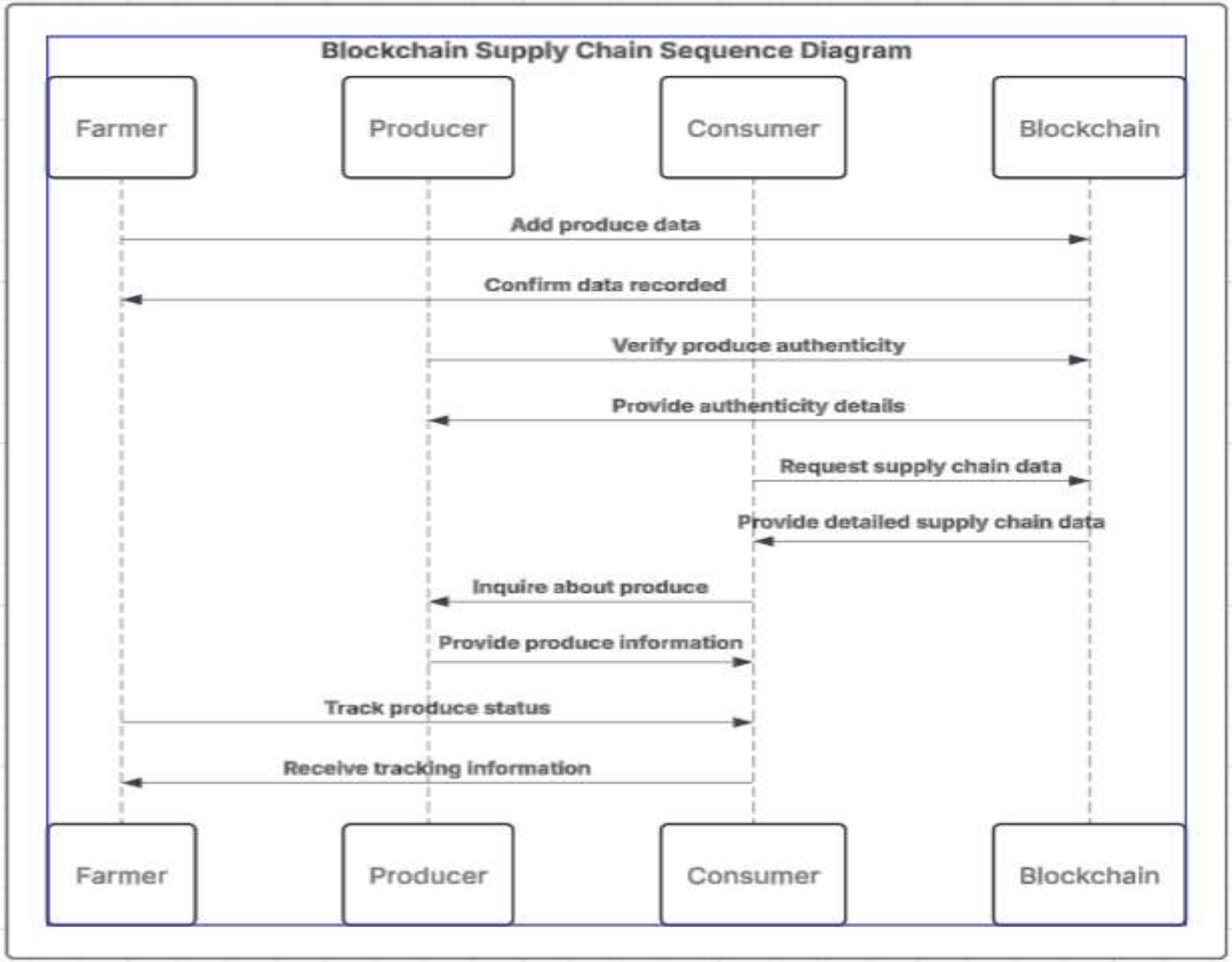
Non-Functional Requirements:

1. **Security :**
   Ensure the system is secure and protects user data.
2. **Scalability :**
   Design the system to scale with increasing user adoption and data volume.
3. **Usability :**
   Create a user-friendly interface for easy navigation.
4. **Performance:**
   Ensure the system responds quickly to user interactions and data queries.

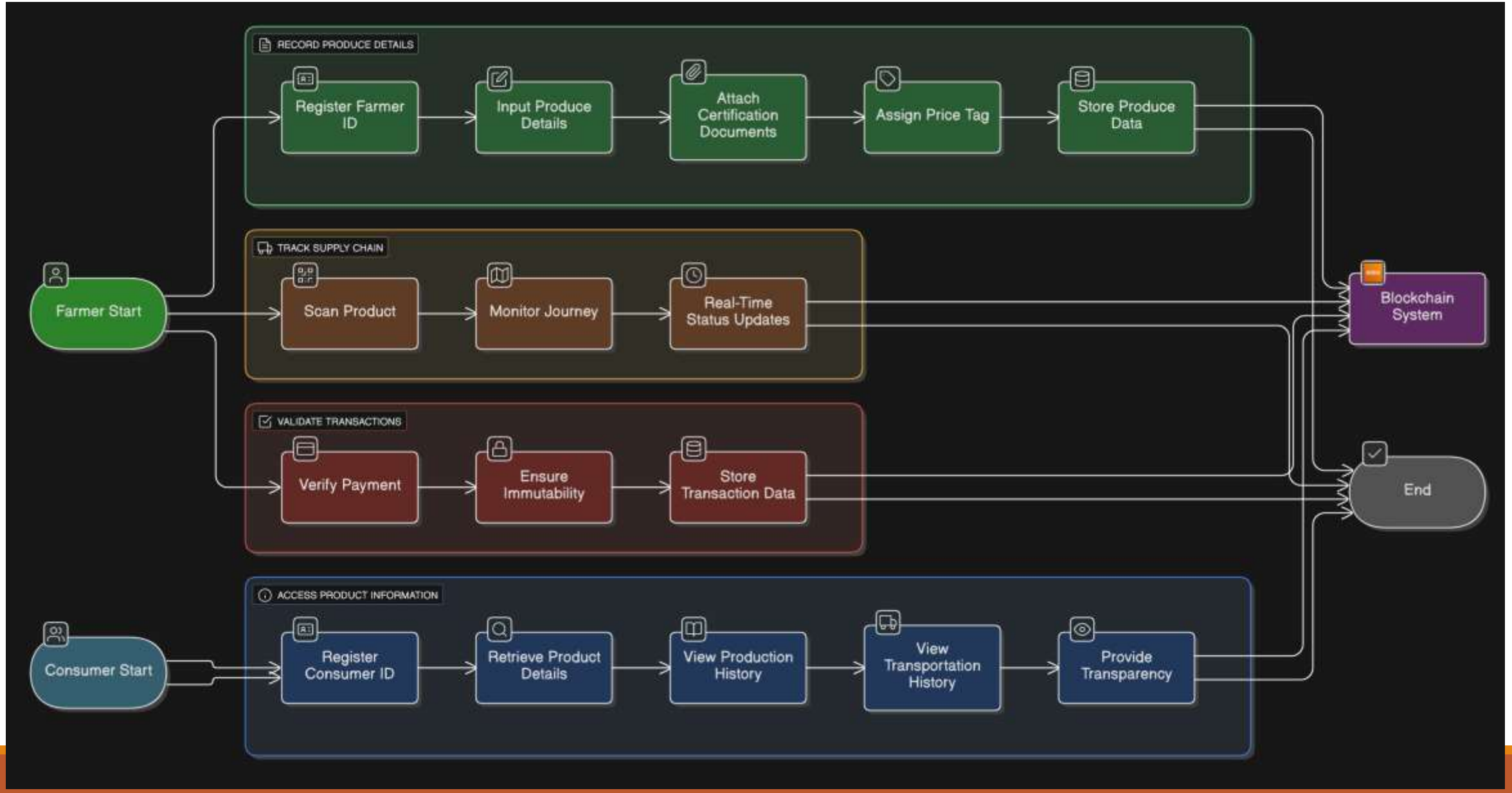
DESCRIPTION OF MODULES:

1. **Product Registration:** Creates a **unique digital ID** for each batch and records initial details (product type, quantity).
2. **Event Logging:** Records time-stamped **events** (harvest, quality checks, inputs used). Each log is a transaction request to the blockchain..
3. **Shipment Management:** Creates the **chain-of-custody** record, linking the product to logistics data. Smart **Contracts**: Automatically executes rules and transactions.
4. **Transaction Validation:** Verifies the integrity of data submitted by the Farmer.(Blockchain)
5. **Product Tracking Interface:** Allows consumers to query the blockchain using a **unique ID** (QR code scan) to retrieve the complete, verified history.
6. **Purchase Recording:** Records the final order transaction, linking the **Order Number** to a unique **Blockchain Transaction Hash**, closing the traceability loop.

UML DIAGRAM


Blockchain Supply Chain Sequence Diagram
Farmer
Producer
Consumer
Blockchain
Add produce data
Confirm data recorded
Verify produce authenticity
Provide authenticity details
Request supply chain data
Provide detailed supply chain data
Inquire about produce
Provide produce information
Track produce status
Receive tracking information
Farmer
Producer
Consumer
Blockchain

USE CASE DIAGRAM:



IMPLEMENTATION :

Login Page- Users (farmer, transporter, warehouse, retailer) log in with their role.

Dashboard- Shows options based on user type.

Add Product Batch (Farmer)- Form to enter: product type, quantity, harvest date, farm location.

Sends data to backend → blockchain.

Update Product Movement (Transporter/Warehouse)

Enter current location, date/time, condition.

Updates blockchain record.

Retailer Page- Confirms product arrival and displays final details.

Consumer Scan Page- User scans QR code → webpage shows the **entire history** from farm to store and can order it.

TESTING:

1. Unit Testing:

include Smart contract functions, API routes, QR code generation, Authentication logic.

2. Integration Testing(Testing how components work together):

include Frontend + backend communication, Backend + blockchain.

3. System Testing:

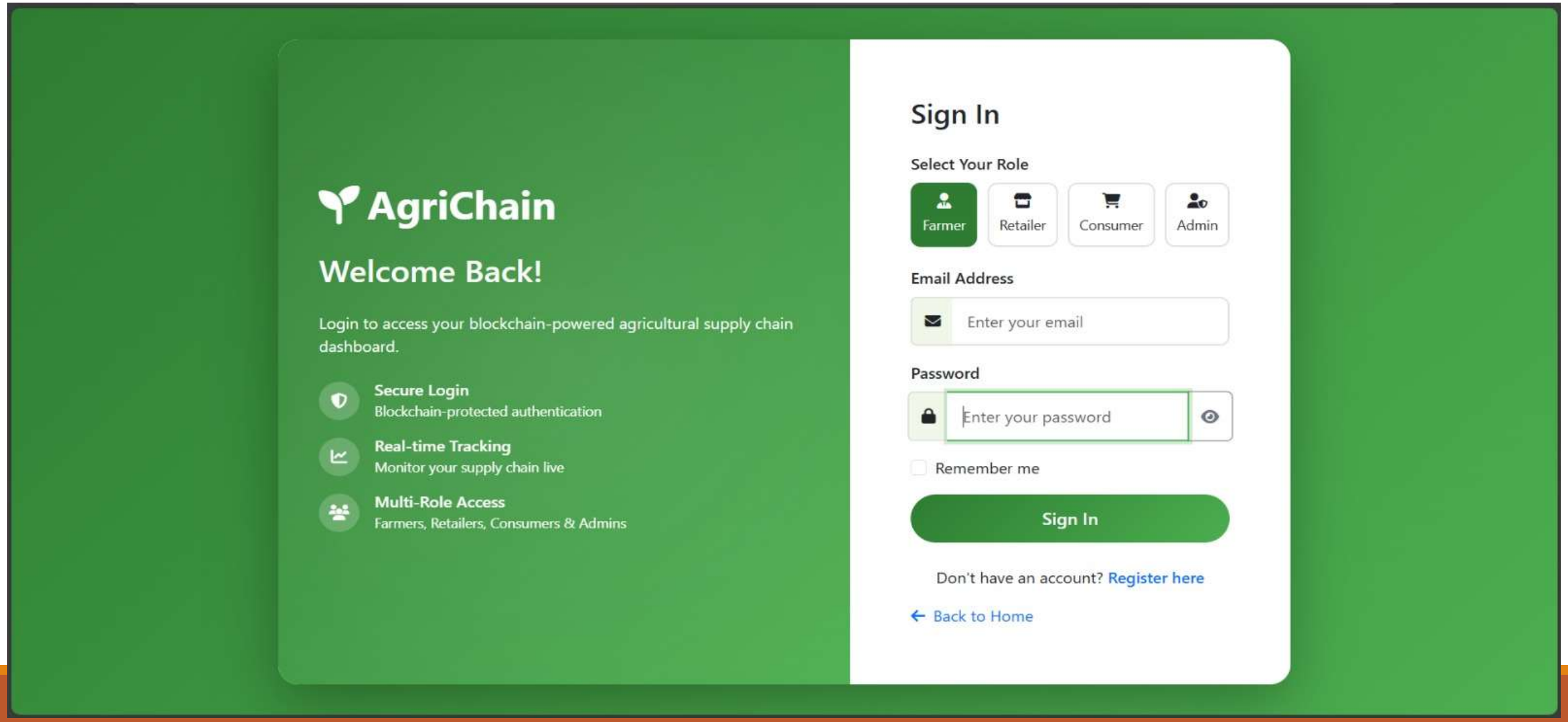
Testing the entire system end-to-end.

4. Performance Testing:

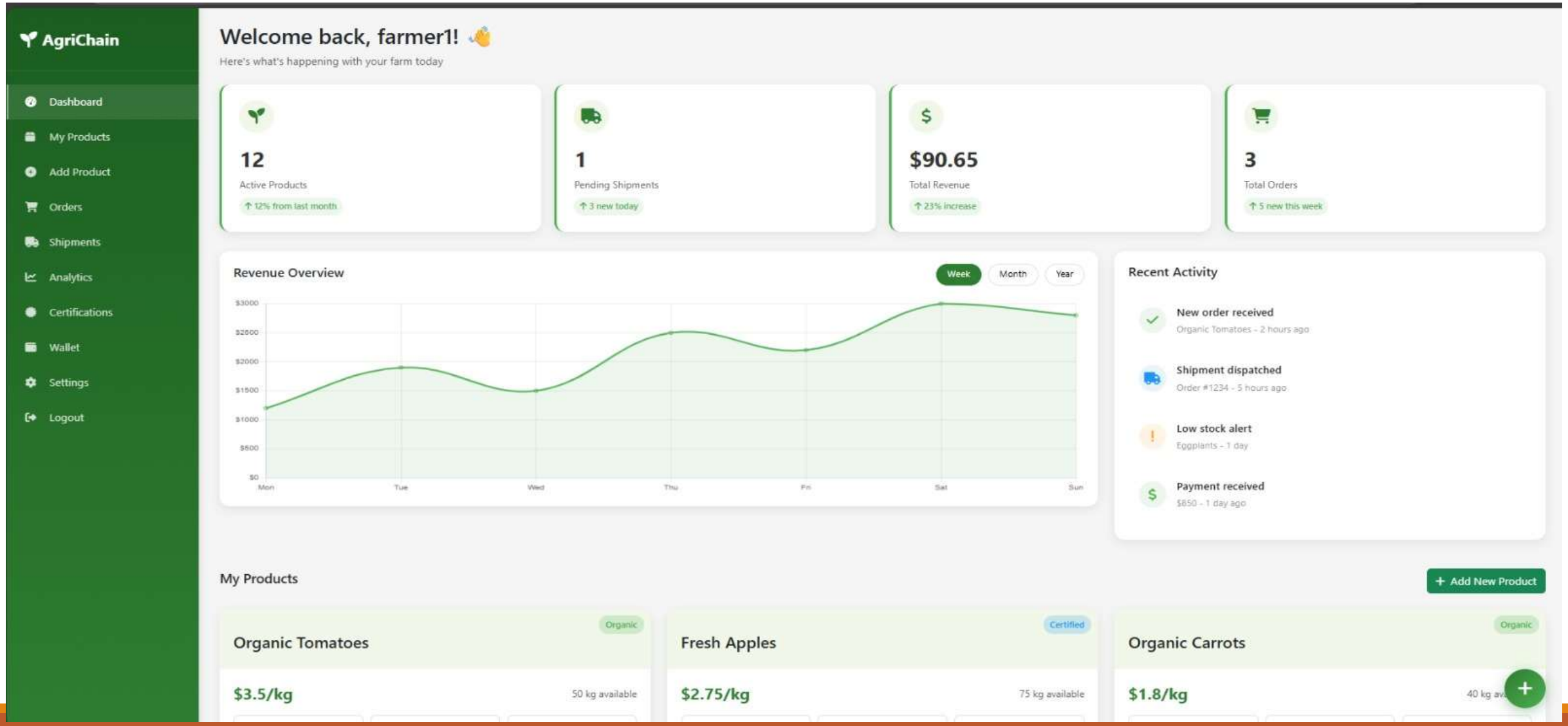
Testing speed, load, and response time.

RESULT

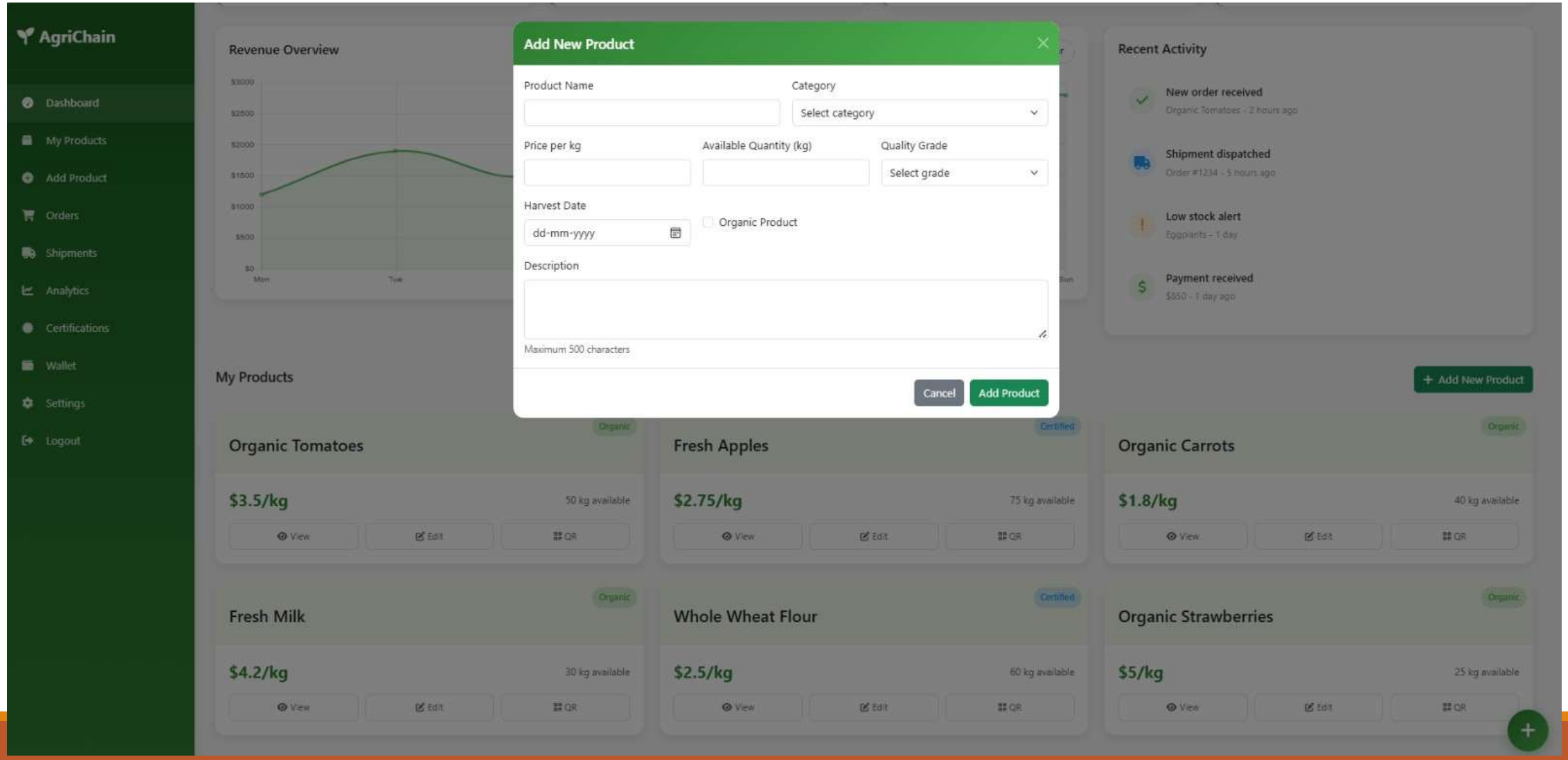
Secure Multi-Role Access: The AgriChain login portal authenticates users (Farmer, Retailer, Consumer, Admin) using blockchain-protected security for real-time supply chain monitoring.



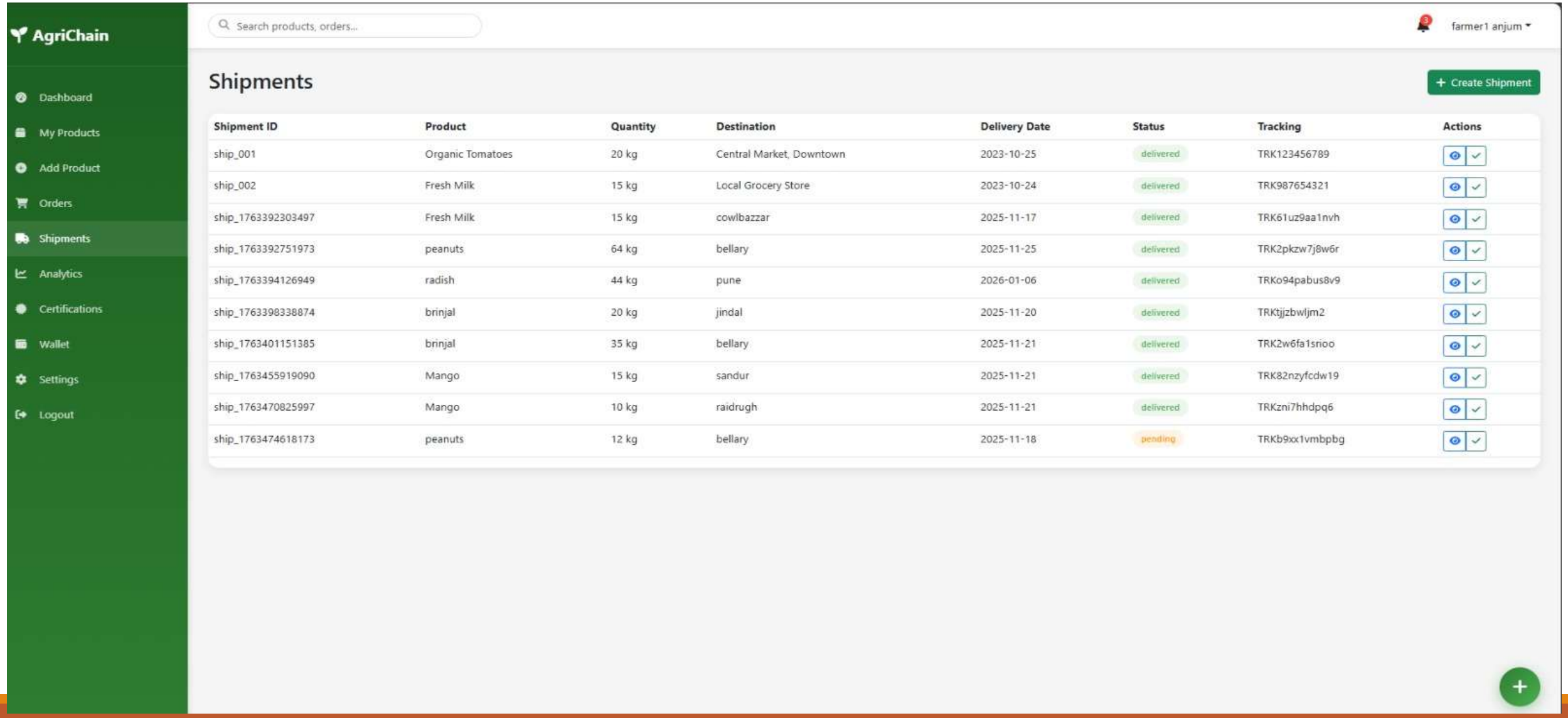
Dashboard: This dashboard provides farmers with key operational metrics, including active products, pending shipments, total revenue, and recent activity alerts.



Product Data Entry (Blockchain Ingestion): The "Add New Product" interface allows farmers to capture and record essential product attributes, harvest details, and quality grades onto the secure ledger.

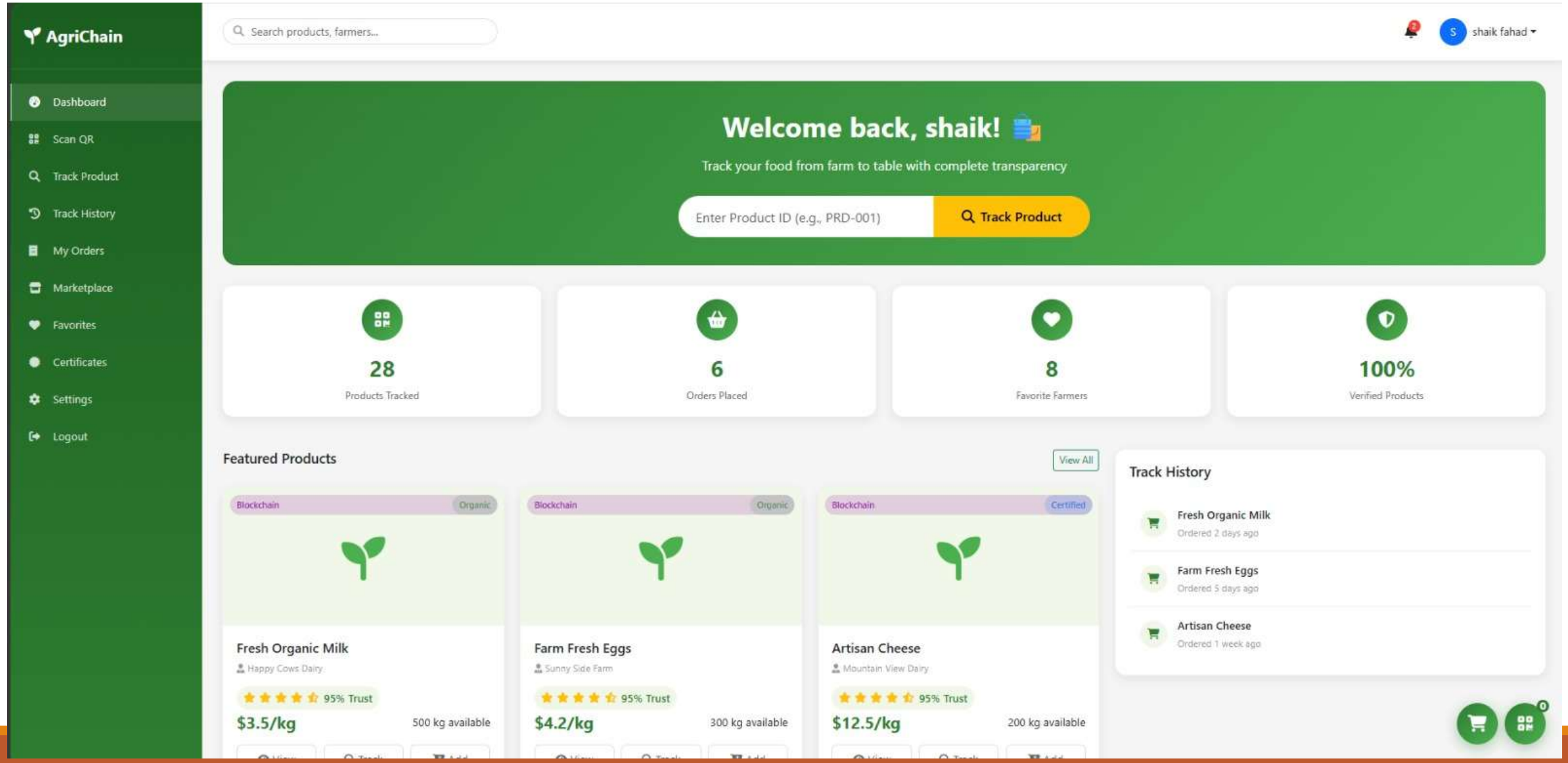


Shipment Traceability: The Shipments log displays the real-time status and unique tracking IDs for all produce deliveries, ensuring transparency from farm to destination.

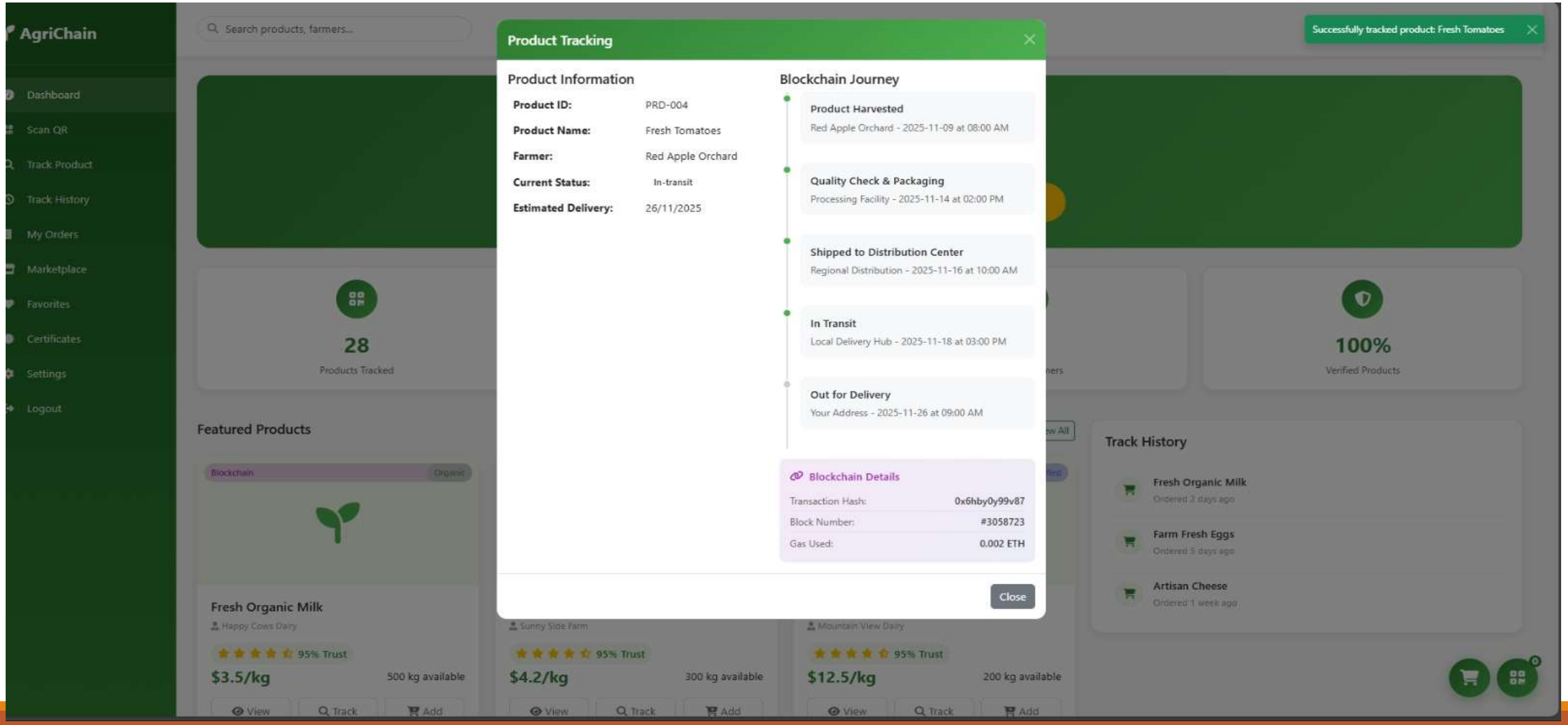


Shipment ID	Product	Quantity	Destination	Delivery Date	Status	Tracking	Actions
ship_001	Organic Tomatoes	20 kg	Central Market, Downtown	2023-10-25	delivered	TRK123456789	
ship_002	Fresh Milk	15 kg	Local Grocery Store	2023-10-24	delivered	TRK987654321	
ship_1763392303497	Fresh Milk	15 kg	cowlbazzar	2025-11-17	delivered	TRK61uz9aa1nvh	
ship_1763392751973	peanuts	64 kg	bellary	2025-11-25	delivered	TRK2pkzw7j8w6r	
ship_1763394126949	radish	44 kg	pune	2026-01-06	delivered	TRKo94pabus8v9	
ship_1763398338874	brinjal	20 kg	jindal	2025-11-20	delivered	TRKtjjzbwljm2	
ship_1763401151385	brinjal	35 kg	bellary	2025-11-21	delivered	TRK2w6fa1srioo	
ship_1763455919090	Mango	15 kg	sandur	2025-11-21	delivered	TRK82nzyfcdw19	
ship_1763470825997	Mango	10 kg	raidrugh	2025-11-21	delivered	TRKzni7hhdpq6	
ship_1763474618173	peanuts	12 kg	bellary	2025-11-18	pending	TRKb9xx1vmbpbg	

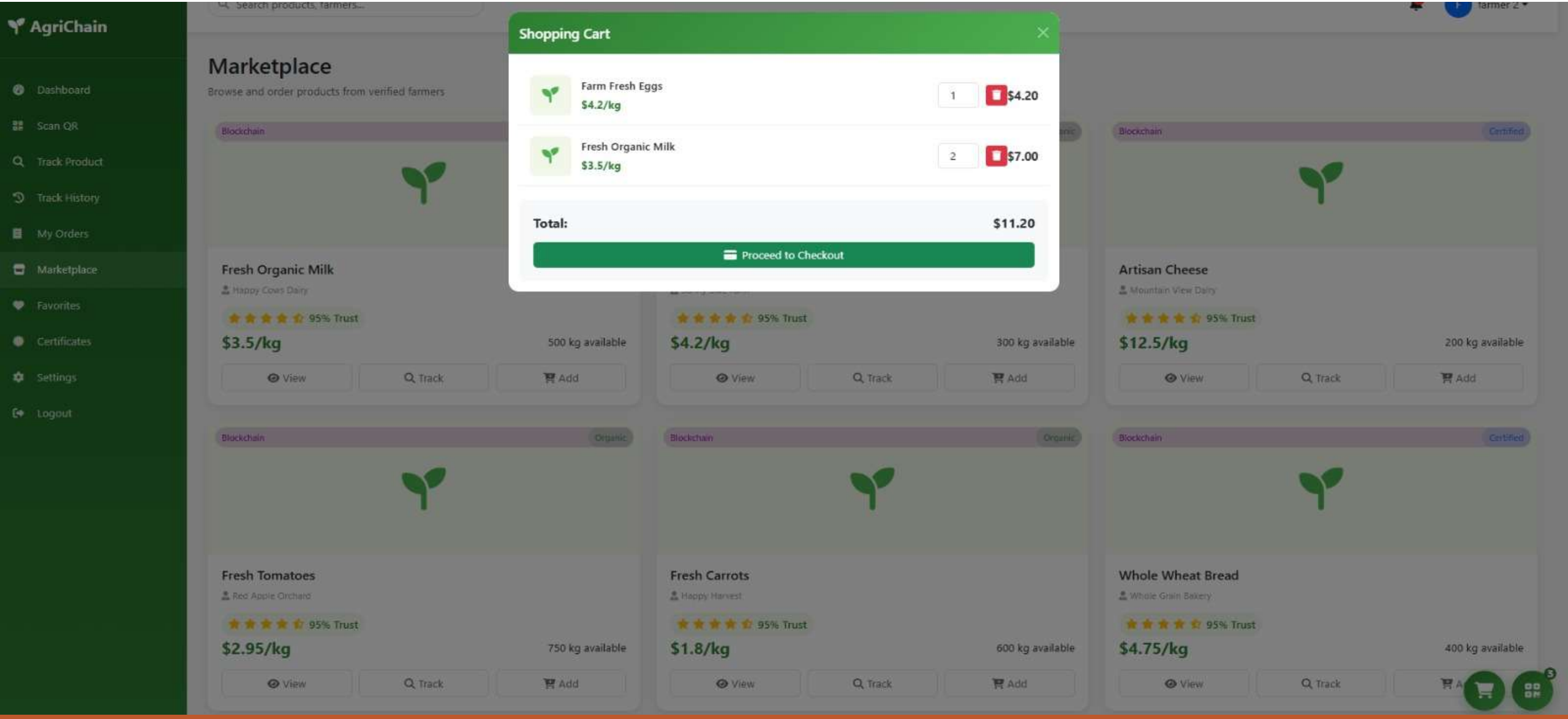
Track Your Food to the Farm: Instantly verify product origin, quality, and compliance from a dashboard of trusted farmers.



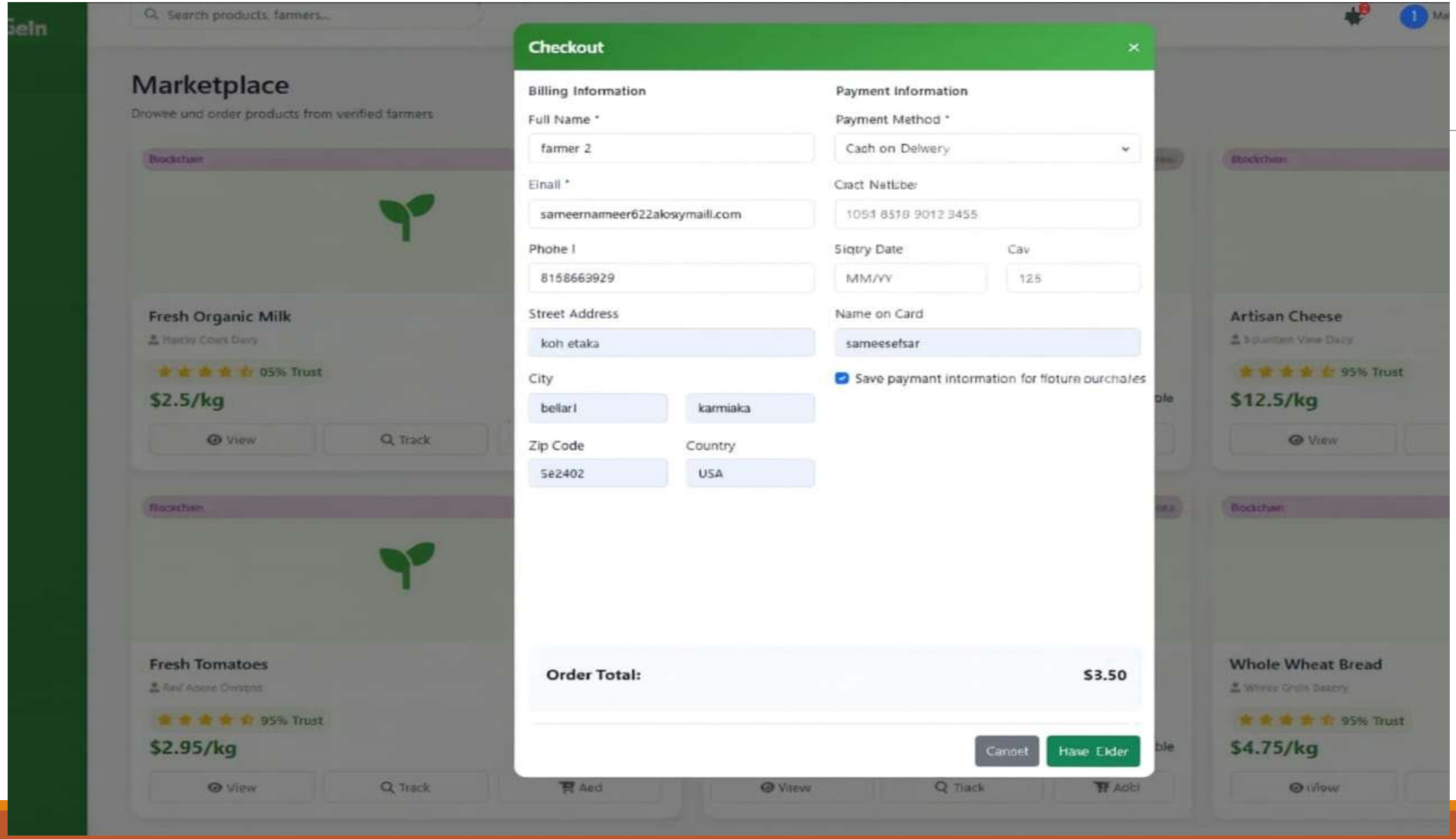
History: View every step of your food's journey, guaranteed by an unchangeable blockchain transaction record



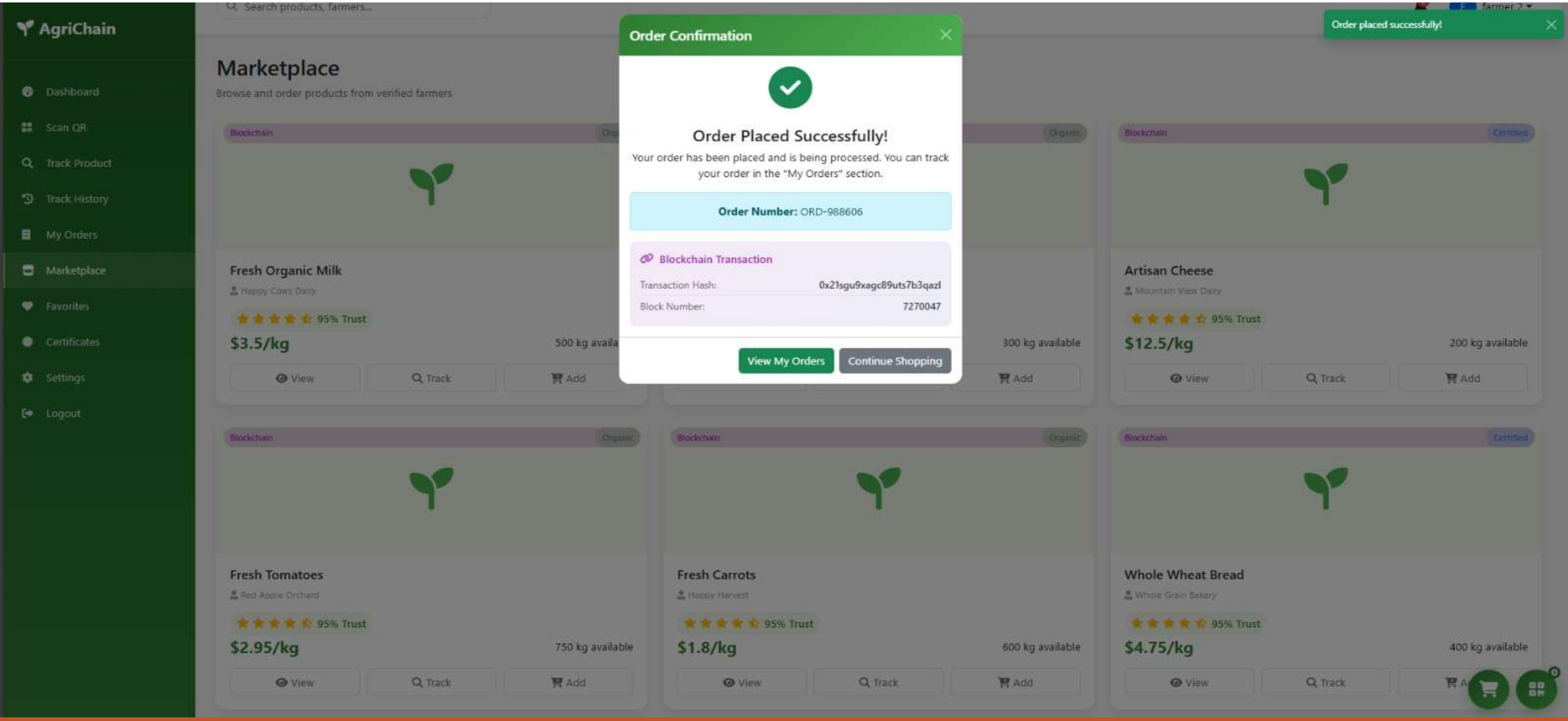
Shop with Certainty: Easily select verified, blockchain-listed goods and proceed to purchase with full price transparency.



Secure Your Order: Fast, standard checkout process for seamless delivery of certified, tracked products.



Purchase on the Chain: Order confirmed and instantly recorded, providing a unique Blockchain Transaction Hash for verifiable ownership.



CONCLUSION:

The successful deployment of AgriTrace not only addresses a technological challenge but also facilitates a major step toward a more **equitable and accountable food system**. By minimizing information asymmetry, the system provides **producers** with verifiable proof of their high-quality, sustainable practices, which can lead to fairer market access and pricing. This project is a robust proof-of-concept, establishing the framework for the **future of transparent food systems**. Future work will focus on scaling this solution across diverse crop types and integrating **smart contracts** to automate key business logic, such as automating payment release to farmers upon verifiable confirmation of goods receipt and quality checks by distribution centers. This will further streamline transactions and solidify AgriTrace as a leading model for agricultural supply chain transparency.

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